GPU Architecture and Its Role in Deep Learning

Zhirui Hu

Department of Statistics

Oregon State University

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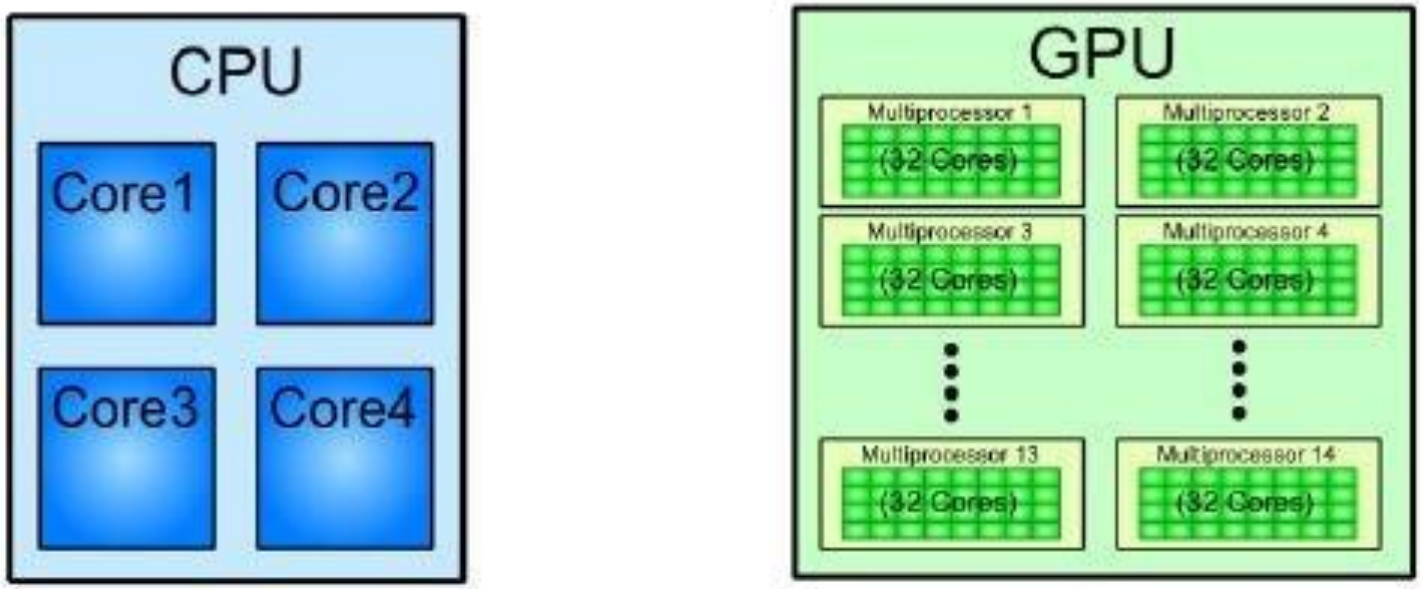
Graphics Processing Unit (GPU)

- A GPU is a specialized electronic circuit designed for rapid mathematical calculations, originally designed for rendering images, videos, and 3D graphics, now used for modern AI development.
- Unlike a CPU's few cores, GPUs have hundreds or thousands of smaller cores designed for parallel processing, execute the *same operation* on many data points simultaneously, called SIMD (Single Instruction, Multiple Data)
- Handle data-intensive tasks like gaming, AI, video editing and scientific simulation.

CPU/GPU Architecture Comparison



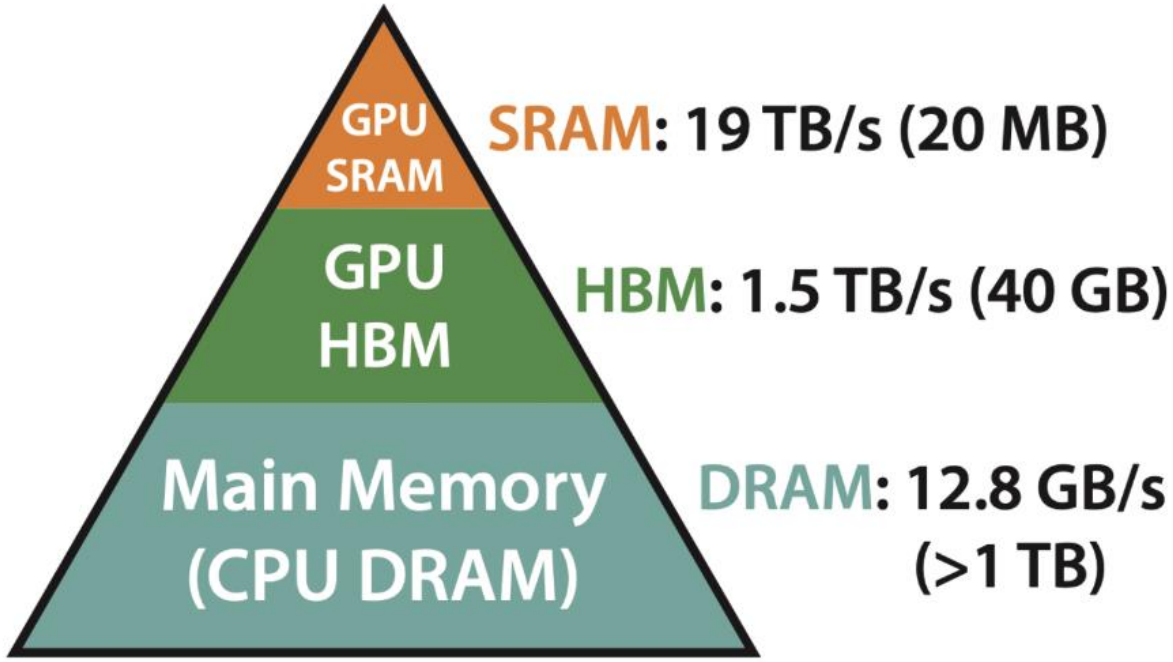
CPU: Has a few powerful cores that process tasks sequentially or with limited parallelism.
GPU: Contains **thousands of smaller cores** designed for **massive parallelism**.

a **CPU is optimized for general-purpose computing**, a **GPU excels in handling thousands of smaller, independent computations simultaneously**.

https://medium.com/ai-insights-cobet/understanding-gpu-architecture-basics-and-key-concepts-40412432812b

GPU memory hierarchy

- **SRAM (Static RAM) — Fastest Cache Memory**

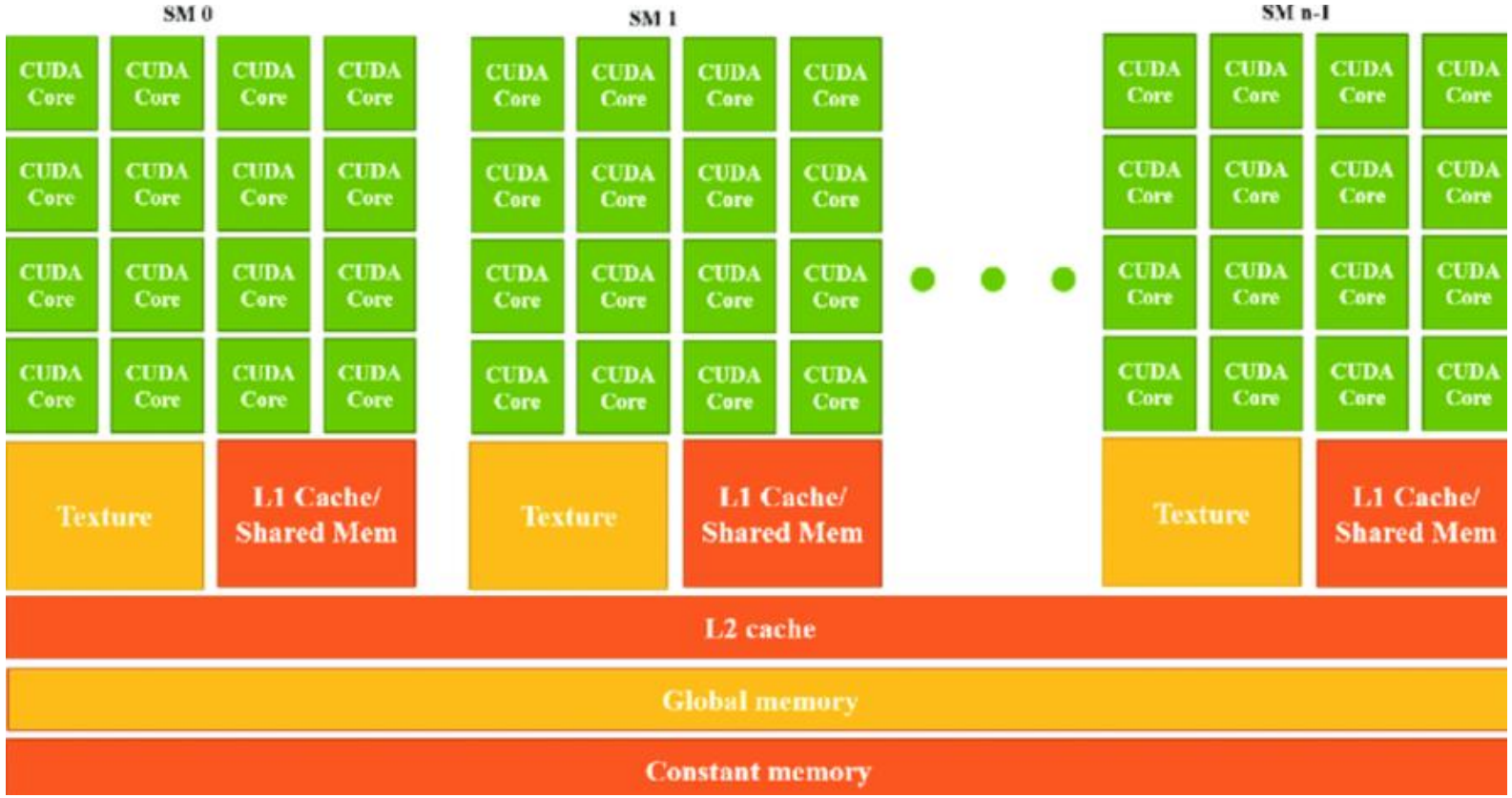


SRAM (Register/L1/L2 cache) helps GPUs process data much faster by reducing the need to fetch data from the slower global memory (VRAM/DRAM).

HBM: High Bandwidth Memory (Global memory) (>> CPU Memory)

Optimizing how data is stored improves GPU performance

Streaming Multiprocessors (SMs)



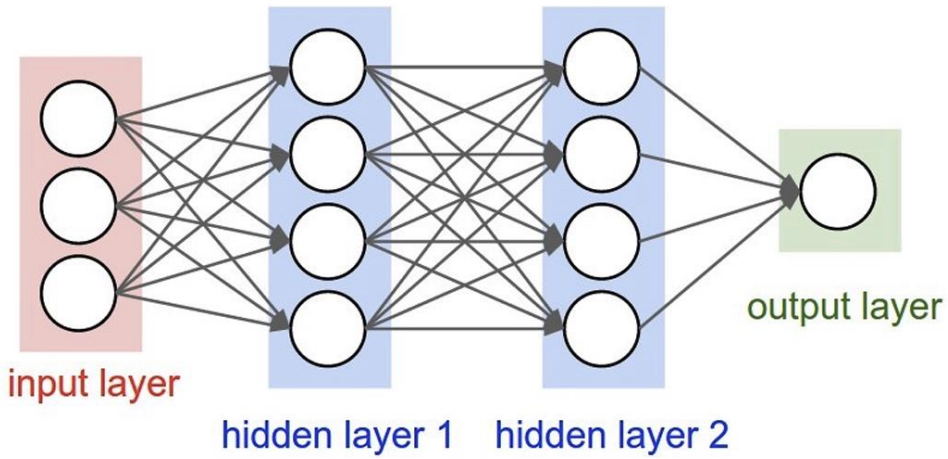
- SM schedules and executes multiple warps
- All cores in a warp must execute the same instruction simultaneously but operate on different data, using multiples of 32 in tensor shapes ensures that all cores in a warp are utilized efficiently
- Each GPU core can perform one Floating Point Operation (FLOP) per cycle.
- Keep memory usage per program low to ensure more programs (PIDs) run per SM

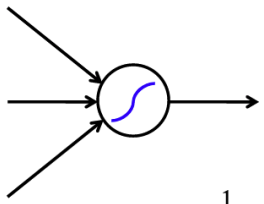
CPU vs GPU runtime in Monte Carlo Simulations

- Estimating π with Random Points
- Bootstrap Confidence Interval

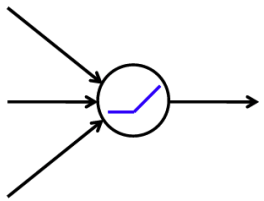
Neural networks

- Perceptron -> Multi-layer perceptrons -> Deep Learning





Sigmoid: $g(t) = \frac{1}{1+e^{-t}}$



Rectified linear unit (ReLU): $g(t) = \max(0,t)$

- Many crucial algorithms and innovations were developed in the 1980's and 1990's, but applications and development were **limited by available computational power.**

Deep Learning Boom & GPUs

- massive datasets, high-performance computing power (GPUs), and advanced neural network architectures
- Matrix multiplies (forward pass, backprop, weight updates) and convolution split into many independent tasks, turning simpler per-core performance into enormous aggregate throughput through massive parallelism.
  - Multiply of two 4096×4096 matrices involves ~34 billion floating-point operations, all of which are *independent* of each other and can run in parallel.
  - Data parallelism

GPU vendors and models

- NVIDIA:
  - **Data Center/High-End:** B200 (Blackwell) (best for massive AI), H200/H100 (standard for training), A100 (40GB/80GB).
  - **Workstation/Value:** RTX 6000 Ada, L40S, RTX A6000.
  - **Consumer/Entry-Level:** RTX 4090, RTX 4080, RTX 3090/3080 Ti.
- AMD:
  - **Data Center:** AMD Instinct MI300X accelerators (used in top AI cloud platforms)
- Google:
  - TPU v4/v5p (optimized for TensorFlow)
- Apple M-Series:
  - M4: up to 80 Apple GPU cores
  - Consumer level: ML inference, no CUDA support

Comparison of NVIDIA GPU models

Model	Arch.	VRAM	CUDA	BW (GB/s)	TDP (W)	FP32 (TF)	Price
RTX 4060	Ada	8 GB GDDR6	3,072	272	115	15.1	$299
RTX 4070	Ada	12 GB GDDR6X	5,888	504	200	29.1	$599
RTX 4080 Super	Ada	16 GB GDDR6X	10,240	736	320	52.2	$999
RTX 4090	Ada	24 GB GDDR6X	16,384	1,008	450	82.6	$1,599
RTX 5090	Blackwell	32 GB GDDR7	21,760	1,792	575	**218**	$1,999
RTX 4000 Ada	Ada	20 GB GDDR6	6,144	432	130	26.7	$1,250
RTX 5000 Ada	Ada	32 GB GDDR6	12,800	576	250	65.3	$4,000
RTX 6000 Ada	Ada	48 GB GDDR6	18,176	960	300	91.1	$6,800
H100 SXM	Hopper	80 GB HBM3	16,896	3,350	700	66.9	$30k
H200 SXM	Hopper	141 GB HBM3e	16,896	**4,800**	700	66.9	$40k
B200 SXM	Blackwell	**192 GB HBM3e**	20,480	**8,000**	1,000	40*	$70k

Consumer Professional Data Center

** B200 FP32 at 40 TF; AI/FP8 throughput is ~4× higher*

TF: TFLOPS, one trillion floating-point calculations per second

▲ Memory Bandwidth Leap

AI inference is bandwidth-bound. Consumer GPUs tops at ~1.8 TB/s (RTX 5090). Data Center B200 reaches 8 TB/s, a 4× jump.

■ VRAM is the ML Bottleneck

Most consumer GPUs cap at 8-24 GB, too small for large models. The RTX 4090's 24 GB makes it the go-to for local ML. Professional GPUs offer up to 48 GB; H200 delivers 141 GB.

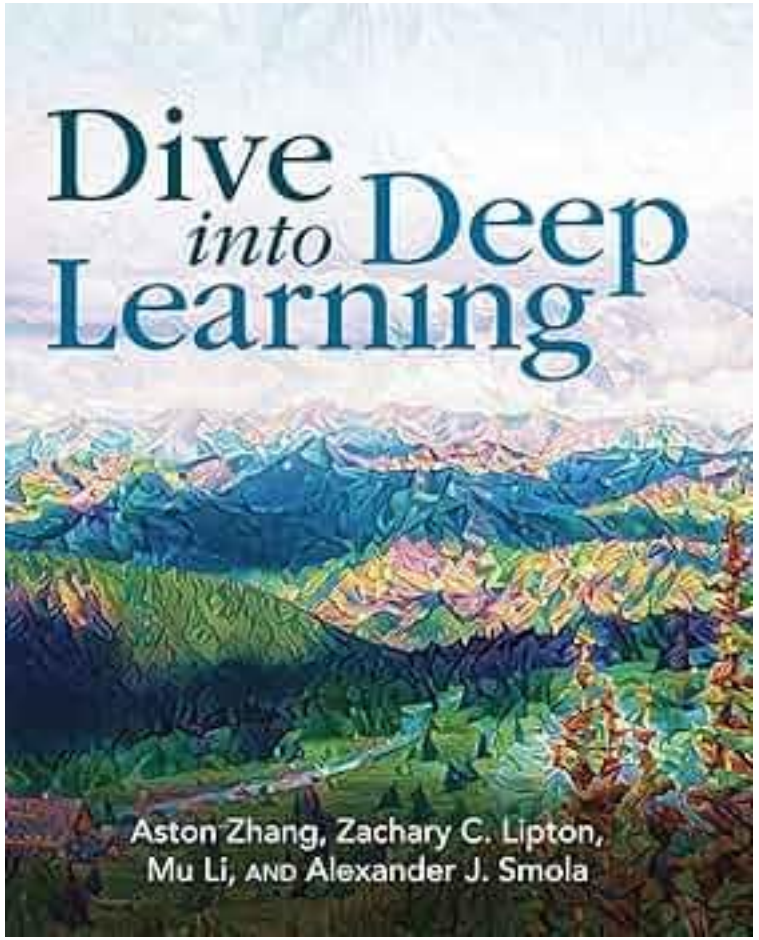
● Architecture Generations

Ada Lovelace (2022) dominates consumer & professional segments. Hopper (2022) powers H100/H200. Blackwell (2024–25) introduces B200 and RTX 5090 with massive gains in FP8 AI throughput.

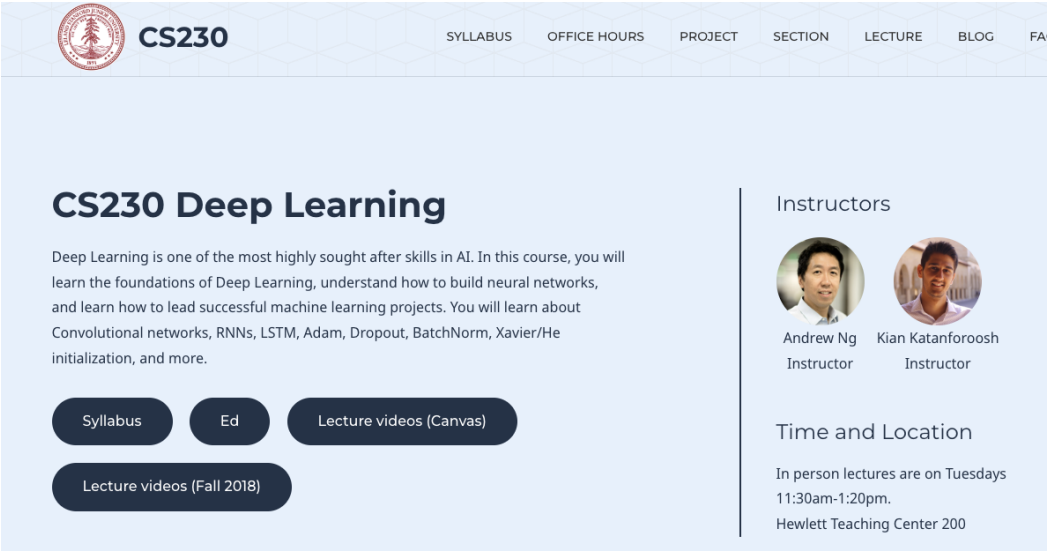
◆ Price vs. Performance Tiers

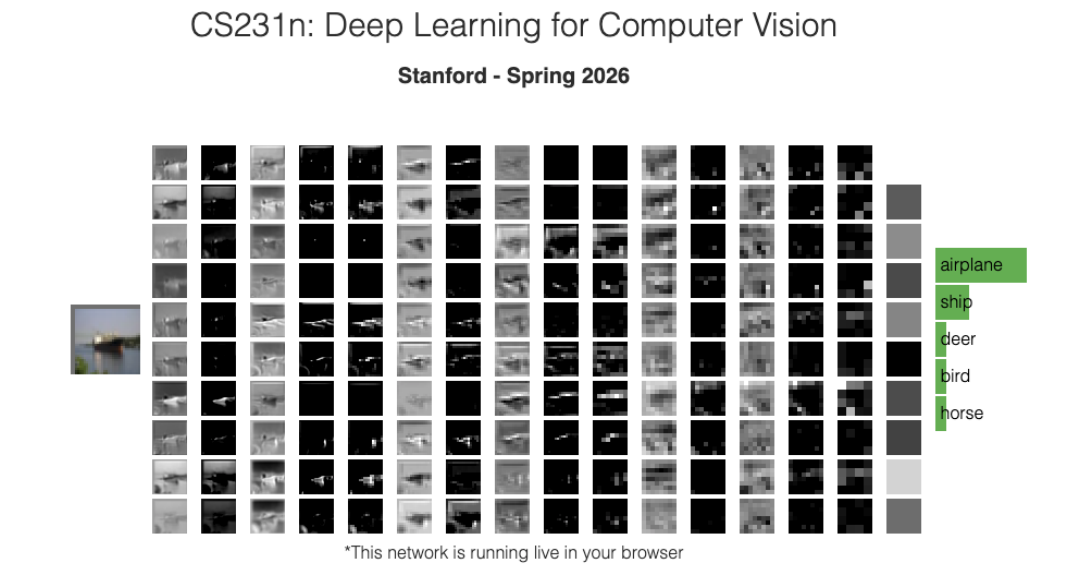
Consumer GPUs offer best $/TFLOP for gaming and light ML. Professional add ECC memory and ISV certifications. Data Center cards cost 15–40× more but deliver unmatched memory capacity and bandwidth.

References



https://www.d2l.ai/chapter_multilayer-perceptrons/mlp.html





http://cs230.stanford.edu/

https://cs231n.github.io/

Course material & Communication

- Github: https://github.com/xyz111131/AI-Tools-for-Statistical-Research

- Slack: https://join.slack.com/t/ai-toolsgroup/shared_invite/zt-3uytyukau-d22XvRJznqomm779lehjtQ

Next lecture

- Sign up for computing cluster access
  - Novus: https://arcs.oregonstate.edu/novus-cluster
  - COE: https://it.engineering.oregonstate.edu/hpc
  - Cisco Secure Client: https://mysupport.oregonstate.edu/esp?id=kb_article_view&sys_kb_id=9576db4883f632d4169bcae0deaad3fa
- Google Colab account
  - https://colab.research.google.com/
  - Free Colab Pro for students and educators